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Stereolithography 3D printing upon near-infrared photopolymerization

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ABSTRACT

This research proposes a novel stereolithography (SLA) three-dimensional (3D) printing strategy based on near-infrared (NIR) photopolymerization through photoinduced electron transfer (PET) between cyanine and iodonium salt, reported here for the first time. In contrast to traditional ultraviolet (UV) photopolymerization, NIR method has garnered significant attention due to its safety and deep penetration capabilities. The NIR-sensitive 3D printing formulas were successfully designed, and various products were fabricated using a customized 3D printer equipped with an 808 nm laser source. Furthermore, due to the weak absorption of coloured fillers in the NIR region compared to cyanine, this strategy can easily manufacture polychromatic products, particularly black, and stimulus-responsive materials capable of photochromism, which is difficult to achieve through 3D printing with UV photopolymerization. The utilization of carbon black (CB) as a light absorber with NIR absorption properties enabled the fabrication of black products and led to an enhancement in printing fidelity. This research introduces a novel processing technique for NIR 3D printing and promotes advancements in the 3D printing field, with significant practical applications.

1. Introduction

Light is truly remarkable. Driven by light, small saplings emerge from the soil, gradually growing into towering trees through continuous illumination. This phenomenon is analogous to the evolving technology of 3D printing based on photopolymerization. This technology utilizes light to induce the curing of liquid materials, allowing them to gradually take shape, much like the growth of "trees." Photopolymerization-based 3D printing has attracted significant attention due to its environmentally friendly attributes, including energy conservation and sustainability.[1–6] This technology typically possesses advantages such as mild processing conditions, a nearly 100 % material utilization rate, and high production efficiency and resolution, which enables it to be widely applied.[1,7–11].

Short-wavelength sources, such as near-UV and blue light, are utilized as the primary sources for 3D photopolymerization printing, as commercially available photoinitiators (PIs), such as acylphosphine oxides and thioxanthones, exhibit photosensitivity to short-wavelength light.[8,12–14] Nevertheless, the utilization of longer wavelength light as a photopolymerization source has emerged as a significant area of interest over the past several years. Short-wavelength light poses health risks to humans, and its limited penetration capability has hindered the development of deep polymerization.[2,15–17].

Consequently, photopolymerization using NIR light as a source (wavelengths greater than 700 nm) has emerged as a promising alternative, as NIR light poses fewer health risks compared to UV light, which is crucial for practical applications concerning human health.[1,18–20] Additionally, the exceptional penetration capability of NIR light offers two significant advantages. Firstly, the curing depth of NIR photopolymerization surpasses that of traditional short-wavelength photopolymerization.[17,19,21–23] Secondly, for systems with coloured fillers, NIR photopolymerization is usually unaffected, as the absorption of coloured fillers in the NIR region is relatively weak.[2,24] This characteristic enables NIR light to penetrate effectively and initiate reactions. In contrast, near-UV and blue light photopolymerization often struggle with this limitation, as the absorption of short-wavelength PIs can be adversely impacted by the presence of coloured fillers, ultimately affecting the polymerization process.[2] Given these advantages, employing NIR light as a photopolymerization source for 3D printing is of considerable significance.

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To date, related studies have been reported. For instance, 3D printing utilizing up-conversion nanoparticles (UCNPs) in direct ink writing (DIW) mode has been successfully employed to manufacture ceramic products under an NIR source.[25–27] Achieving fast 3D printing with lower energy laser using triplet fusion up-conversion nanocapsules (UCNCs).[28] Photoinduced free radical promoted cationic reversible addition–fragmentation chain transfer (RAFT) polymerization for 3D Printing.[29–31] Additionally, 3D printing has been conducted via digital light processing (DLP) using cyanine as NIR photosensitizers.[32] A brief summary of this works is shown in Table S1. Undoubtedly, these studies are pioneering and have achieved remarkable results. However, research on NIR 3D printing remains extremely limited, and some methods still have room for improvement. Applying relatively lower-intensity light to save energy, utilizing more accessible raw materials, and manufacturing more complex, precise, and larger products are research aims that should be explored. Therefore, there is an urgent need to develop a novel NIR 3D printing method to address these challenges.

We designed a novel 3D printing method for the first time in SLA mode via NIR photopolymerization based on a photoinitiating system (PIS) composed of cyanine and iodonium salt. Here, cyanine serves as a photosensitizer that absorbs NIR light and undergoes a PET reaction with iodonium salt to initiate polymerization.[33–39] The design and screening process of the photopolymerization formulas was provided, followed by actual 3D printing tests that yielded exciting results. The advantages of NIR light were utilized to print the polychromatic and stimulus-responsive products. Notably, the successful manufacture of black products is extremely meaningful, as it is challenging to prepare them through near-UV 3D printing. The addition of black fillers can not only colour the products but also function as a light absorber to improve printing precision; this strategy can achieve two objectives with one solution. Our research provides a feasible strategy for the design of 3D printing formulas and enriches 3D printing methods by employing NIR light as source, which is significant for the development of 3D printing.

2. Experiment section

2.1. Materials

The cyanines 1-butyl-2-((*E*)-2-((*E*)-3-(2-((*E*)-1-butyl-3,3-dimethylindolin-2-ylidene)ethylidene)-2-(phenylthio)cyclohex-1-en-1-yl)vinyl)-3,3-dimethyl-3*H*-indol-1-ium tetraphenylborate ($1aZ_1X_1$), 1-butyl-2-((*E*)-2-((*E*)-3-(2-((*E*)-1-butyl-3,3-dimethylindolin-2-ylidene)ethylidene)-2-(diphenylamino)cyclopent-1-en-1-yl)vinyl)-3,3-dimethyl-3*H*-indol-1-ium tetrafluoroborate ($2aZ_2X_2$), 1-butyl-2-((1*E*,3*E*,5*E*)-7-((*E*)-1-butyl-3,3-dimethylindolin-2-ylidene)hepta-1,3,5-trien-1-yl)-3,3-dimethyl-3*H*-indol-1-ium hexafluorophosphate ($3aX_3$), 5-((*E*)-6-((*E*)-2-(3-ethyl-1,1-dimethyl-1,3-dihydro-2*H*-benzo[e]indol-2-ylidene)ethylidene)-2-((*E*)-2-(3-ethyl-1,1-dimethyl-1*H*-benzo[e]indol-3-ium-2-yl)vinyl)cyclohex-1-en-1-yl)-1,3-dimethyl-2,6-dioxo-1,2,3,6-tetrahydropyrimidin-4-olate ($4bZ_3$), 2-((*E*)-2-((*E*)-2-(diphenylamino)-3-(2-((*E*)-1-(2-methoxyethyl)-3,3-dimethylindolin-2-ylidene)ethylidene)cyclopent-1-en-1-yl)vinyl)-1-(2-methoxyethyl)-3,3-dimethyl-3*H*-indol-1-ium hexafluorophosphate ($2cZ_2X_3$), and 1,3,3-trimethyl-2-((1*E*,3*E*)-5-((*E*)-1,3,3-trimethylindolin-2-ylidene)penta-1,3-dien-1-yl)–3*H*-indol-1-ium tetrafluoroborate (S0957) were obtained from FEW Chemicals GmbH. (*E*)-*N*-Ethyl-*N*-(4-(1,5,5-tris(4-(diethylamino)phenyl)penta-2,4-dien-1-ylidene)cyclohexa-2,5-dien-1-ylidene)ethanaminium 4-methylbenzenesulfonate ($5X_4$) was obtained from Shanghai Showa Chemicals. 4-Acryloylmorpholine (M1), isobornyl acrylate (M2), 5-ethyl-1,3-dioxan-5-yl) methyl acrylate (M3), 2-(2-ethoxyethoxy)ethyl acrylate (M4), tetrahydrofurfuryl acrylate (M5), sodium tetraphenylborate (STPB), triphenylphosphine (TPP), tetrabutylammonium hexafluorophosphate, ferrocene, perylene, Rhodamine B, 1,3,3-trimethyl-6-nitroindolene-2-spiro-2-benzopyran (SP), and CB were obtained from Beijing Innochem Chemical. The oligomer difunctional polyurethane acrylate LuCure8091 (O1) was acquired from Guangzhou Wraio Chemicals. The oligomer *hexa*-functional aromatic polyurethane acrylate 220 (O2) was obtained from Allnex Resin (China). The spin trapping reagent *N-tert*-butyl-2-phenylnitrone (PBN) was obtained from Shanghai Bide Pharmatech. Bis(4-methylphenyl)iodonium hexafluorophosphate (Iod) and 1-[4-(phenylthio)phenyl]-1,2-octanedione 2-(*O*-benzoyloxime) (OXE-01) were supplied by Changzhou Powerful Photoelectric Materials. Anhydrous acetonitrile, dichloromethane (CH_2Cl_2), methanol and ethanol were supplied by Beijing Chemical Work.

2.2. Characterization

The absorption properties of cyanines and coloured fillers, and the steady-state photolysis behaviour of the 3D printing formula were recorded with a Shimadzu UV-3600 UV − vis − NIR spectrophotometer. The SLA 3D printing equipment was customized by Wuhan Doit Technology Co., Ltd, and the laser with an emission wavelength of 808 nm was used as the light source, and the maximum intensity reached 850 mW cm^{-1}. The liquid chromatography-mass spectrometry (LC-MS) analysis was conducted utilizing an Agilent 1260 high-performance liquid chromatography system coupled with a Bruker micrOTOF-QII-quadrupole time-of-flight tandem mass spectrometer. The printing products were observed via an optical microscope (12XB-PC, Shanghai Optical Instrument Factory) at a magnification of 5×.

2.3. Radiation sources

An MDL-H-808–5 W NIR laser source produced by Changchun New Industries Optoelectronics Tech was utilized as the irradiation device and operated with an emission wavelength of 808 nm. Additionally, an 820 nm light-emitting diode (LED) source (UVEC-60X20, Shenzhen Lamplic) was also employed as an irradiation device. An LED bulb (UVSP-81 T, ShangHai Futansi) that emits at a concentrated wavelength of 365 nm and 580 nm was also used as an irradiation device.

2.4. Redox potentials

The oxidation potentials of all cyanines except $2cZ_2X_3$ and the reduction potentials of Iod and OXE-01 were obtained from the literature, as shown in Table S2. The oxidation potential of $2cZ_2X_3$ was investigated with a CHI660C electrochemical workstation (Shanghai Chenhua). Tetrabutylammonium hexafluorophosphate in anhydrous acetonitrile (0.1 mol L^{-1}) was employed as a supporting electrolyte, and ferrocene was utilized as an external standard. Anhydrous acetonitrile solutions of $2cZ_2X_3$ (1×10^{-3} mol L^{-1}) were tested by adopting a platinum disk as the working electrode and Ag/AgCl as the reference electrode at a scanning rate of 0.1 V s^{-1}. The free energy of photoinduced electron transfer (ΔG_{PET}) was calculated via the Rehm–Weller equation:

$$\Delta G_{PET} = E_{ox} - E_{red} - E_e + C$$

where E_{ox} represents the oxidation potential of cyanines. E_{red} represents the reduction potentials of Iod and OXE-01, E_e represents the energy of the excited state for cyanines (calculated by the maximum absorption wavelength), and the electrostatic interaction (C) is generally negligible in polar solvents.

2.5. Computational procedure

The theoretical calculations of cyanines, Iod, OXE-01, and monomers were performed via Gaussian 09 W software, which is based on density function theory (DFT). The frontier molecular orbits and energies of the cyanines (cationic moieties) and OXE-01 were analysed by optimizing the molecular geometries and frequency calculations at the B3LYP/6-31G* level. The frontier molecular orbitals and energies of Iod (cationic moiety) were analysed by optimizing the molecular geometries and performing frequency calculations at the UB3LYP/LANL2DZ level.[40] The atomic point charges of the monomers and *p*-methylphenyl

radical were analysed by optimizing the molecular geometries and frequency calculations at the B3LYP/6-31G* and UB3LYP/6-31G* levels, respectively, and then calculated via the atomic dipole moment-corrected Hirshfeld (ADCH) method with Multiwfn software.[41] Gaussview 5.0, Multiwfn and VMD software were used for drawing and visual analysis.

2.6. Selection of 3D printing formulas

Various formulas were prepared at cyanines:initiator:STPB:TPP:monomer:O1:O2 mass ratios of 0.001:0.02:0.002:0.002:0.8:0.12:0.08. Among them, the six cyanines included $1aZ_1X_1$, $2aZ_2X_2$, $3aX_3$, $4bZ_3$, $2cZ_2X_3$ and $5X_4$; the two initiators included Iod and OXE-01; and the five monomers included M1, M2, M3, M4 and M5 for selection. These materials were utilized for the screening experiments. The selected optimal formulation was used for actual 3D printing.

2.7. Photopolymerization experiments

The photopolymerization kinetics experiments were carried out with the formulas presented in Section 2.6 via a Nicolet 5700 FT-IR spectroscope. The photosensitive formulas were injected into a laminate with a thickness of approximately 30 μm between two KBr plates and irradiated with a laser at 808 nm (250 mW cm^{-2}). The conversions of the C=C bond were calculated by comparing the peak areas at approximately 820 − 773 cm^{-1} or 1660 − 1600 cm^{-1}.[42,43].

Furthermore, the 3D printing formula was composed of $4bZ_3$, Iod, STPB, TPP, M1, O1 and O2 (0.001:0.02:0.002:0.002:0.8:0.12:0.08:0.02, mass ratio) with addition of OXE-01 (0.02, mass ratio) or CB (0.002, mass ratio). These formulas were tested via the same method described above.

In addition, the 3D printing formula was composed of $4bZ_3$, Iod, STPB, TPP, M1, O1, O2 (0.0002:0.02:0.002:0.002:0.8:0.12:0.08, mass ratio), and coloured fillers including perylene (0.001, mass ratio), Rhodamine B (0.005, mass ratio), S0957 (0.0004, mass ratio), and SP (0.0004, mass ratio). These formulas were tested via the same method described above.

2.8. Thermal imaging

An infrared thermal imaging camera (Testo 890) was employed to monitor the temperature of the photopolymerization experiments and the actual 3D printing process. For the photopolymerization experiments, the formulas and light intensities of the irradiation sources were kept consistent with those of Sections 2.6 and 2.7. These formulas were sandwiched between two transparent glass slides (approximately 30 μm thick) at 25 ± 2 °C. The temperature variations of the sandwich during 50 s of irradiation were recorded.

For the actual 3D printing process, the camera was fixed next to the 3D printing equipment to monitor the temperature variations during the printing process. The recorded videos are presented in supplementary movies 3 and 4.

2.9. Thermal stability tests

The thermal stability of the formulas (as in Sections 2.6 and 2.7) was evaluated via differential scanning calorimetry (DSC) (METTLER TOLEDO Company DSC1). Approximately 4–6 mg of the formulas was heated from 25 °C to 250 °C at a rate of 10 °C min^{-1} under a N_2 atmosphere. The initial thermal reaction temperature (T_i) was determined by identifying the onset of the first exothermic peak.

2.10. 3D printing process

The formula was processed through 3D printing with a light intensity of 145 mW cm^{-2} or 220 mW cm^{-2}. Recoded videos are presented in supplementary movies 1 and 2. After printing was finished, the support structures were disassembled and subsequently subjected to postcuring treatment through irradiation with an 820 nm LED source (825 mW cm^{-2}) for 10 min and placed in a vacuum oven at 60 °C for 1 h. The areas where the support structures were removed were subsequently polished with abrasive paper. Finally, these samples were soaked in ethanol and subjected to ultrasonic cleaning for 10 min to obtain the final products.

2.11. Steady-state photolysis

The steady-state photolysis tests were conducted by preparing a formula composed of $4bZ_3$, Iod, STPB, TPP, M1, O1 and O2 with the same mass ratios as those in Section 2.6. The photosensitive formulation were sandwiched between two quartz slides, each measuring 45 mm in length, 12 mm in width, and 1 mm in thickness. The thickness of the formula layer was approximately 30 μm. The intensities of irradiation were set at 825 mW cm^{-2} for the 820 nm LED.

Furthermore, the formula composed of $4bZ_3$, Iod, STPB, TPP, M1, O1, O2 and $1aZ_1X_1$ (0.0002:0.02:0.002:0.002:0.8:0.12:0.08:0.0001) was tested for the steady-state photolysis behavior by employing the same test method as mentioned above.

2.12. Electron spin resonance (ESR) tests

ESR tests were conducted on the $4bZ_3$/Iod system and Iod alone via a JEOL JES-FA200 spectrometer (X-band). The tests were performed at field modulations of 9.06 GHz and 100 kHz, with a microwave power of 0.998 mW. Free radicals were generated by the $4bZ_3$/Iod system under irradiation with an 820 nm LED (825 mW cm^{-2}) at room temperature and then trapped by PBN. The molar ratio of the $4bZ_3$/Iod/PBN systems was 0.1:0.5:1. The system was dissolved in dichloromethane at a $4bZ_3$ concentration 1×10^{-4} mol L^{-1}.[2].

The free radicals generated by Iod were subsequently trapped by PBN at room temperature through irradiation with a 365 nm LED (80 mW cm^{-2}). The molar ratio of the Iod/PBN systems was 0.5:1. These systems were dissolved in dichloromethane, with an Iod concentration of 2×10^{-4} mol L^{-1}.[2].

2.13. Tensile tests

Tensile tests were conducted on the dogbone samples fabricated via 3D printing via electromechanical universal testing machines (Instron 5969) with a tensile rate of 15 mm min^{-1}. The 3D model of the dogbone is depicted in Fig. S13, and the printing objects presented in Fig. 3c and Fig. S30 (thicknesses of approximately 2.8 mm, widths of approximately 2.6 mm, and lengths of approximately 21 mm) were fabricated simultaneously.

3. Results and discussion

3.1. Design and screening of 3D printing formulas

Several factors must be considered when designing 3D printing formulas, including polymerization rate, volume shrinkage, viscosity, oxygen inhibition, and polymerization temperatures. For the PISs, six cyanines with different structural characteristics and two initiators were employed for screening (Fig. 1a). All cyanines show remarkable NIR absorption capacities (Fig. S1), ensuring sufficient absorption of the light emitted by the 808 nm laser of the printing equipment. Among the heptamethine cyanines, $1aZ_1X_1$ and $4bZ_3$ feature six-membered bridge chain structures, whereas $2aZ_2X_2$ and $2cZ_2X_3$ contain five-membered bridges, and $3aX_3$ has an open-chain moiety. Compared with the previous five structures, $5X_4$ is special, as it has a pentamethine structure but exhibits equally outstanding absorption and photochemical properties.[2,44] Iod and the oxime ester OXE-01 were selected as initiators, as shown in Fig. 1b. This selection is supported by earlier studies

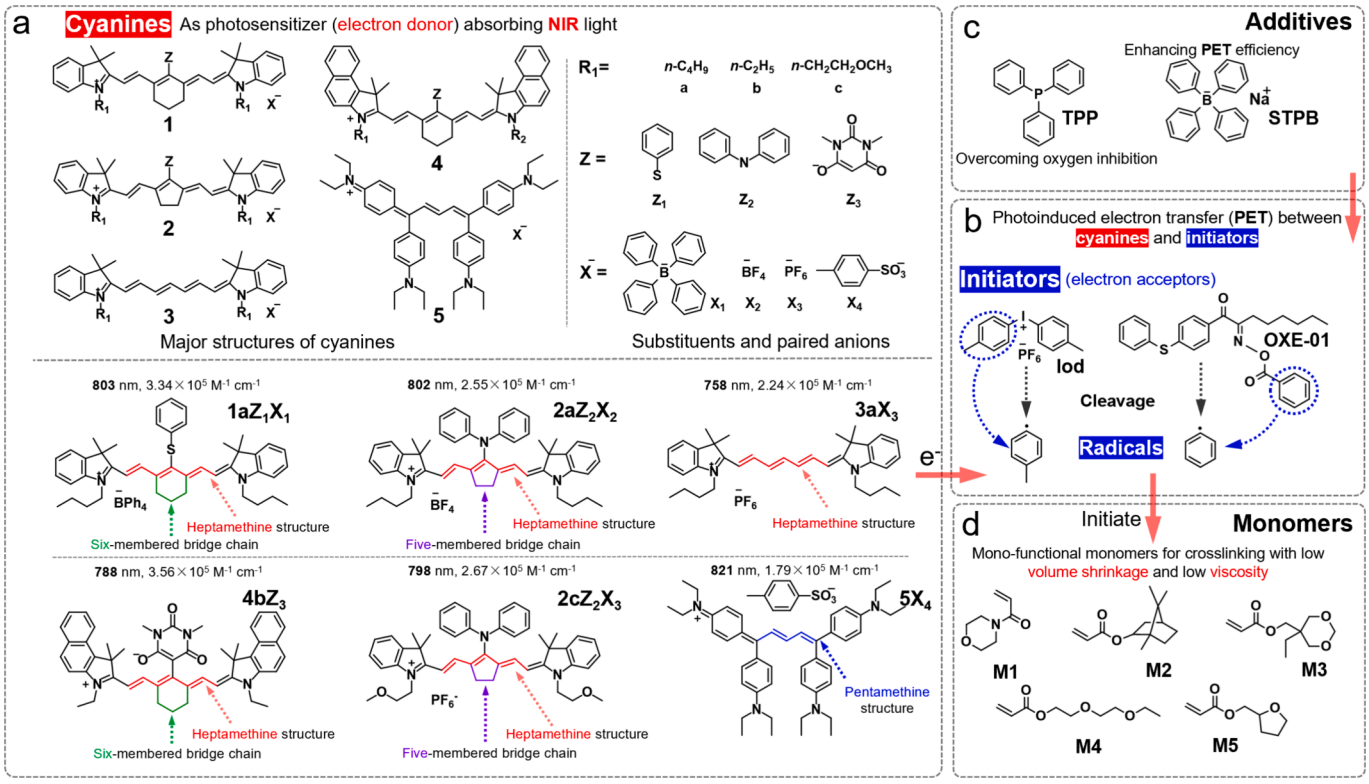


Fig. 1. (a) Chemical structures of six cyanines with distinct structural characteristics that absorb NIR light for the screening of 3D printing formulas. (b) Chemical structures of initiators, including Iod and OXE-01, can undergo PET reactions with cyanines to generate radicals with initiating activity. Chemical structures of (c) additives and (d) monomers used in experiments; among them, five mono-functional monomers are also offered for the screening of 3D printing formulas, while TPP and STPB are utilized as additives to enhance the polymerization performance.

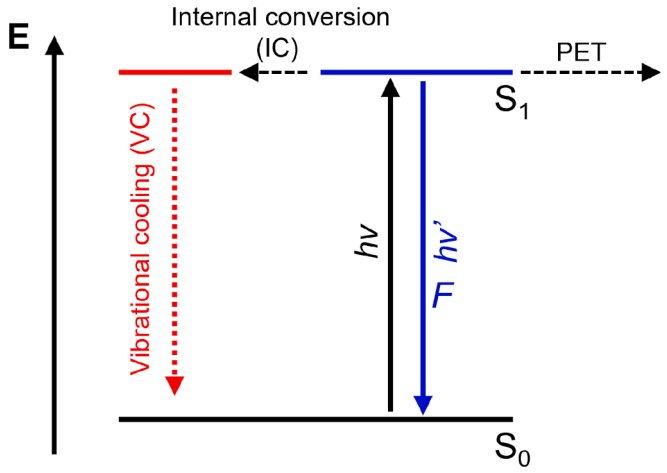
indicating their capability to initiate photopolymerization through PET under NIR light irradiation.[37,39,45] Furthermore, STPB is worthy of use to increase the PET efficiency between cyanines and initiators.[2,36] The addition of TPP is beneficial for overcoming oxygen inhibition because the printing process involves exposure to air, as shown in Fig. 1c.[46] Regarding the polymerizable monomers and oligomers, mono-functional monomers are essential for minimizing volume shrinkage and viscosity. Hence, M1, M2, M3, M4, and M5 were selected (Fig. 1d). To further enhance the polymerization rate and the mechanical properties of the resulting materials, two acrylate oligomers, O1 and O2, were used.

The optimal proportions of each component necessitate careful consideration. For cyanine, given their obvious photothermal conversion effect,[47,48] excess cyanine is bound to drastically increase the temperature of polymerization system, which is unfavourable for the printing process due to uncontrollable thermal polymerization. Conversely, if the cyanine content is overly low, it can facilitate unregulated penetration of NIR light within the formula, leading to a reduction in fidelity. It is a balancing that requires taking both aspects into account simultaneously. Therefore, the mass ratio of cyanine to the polymerizable components (monomer and oligomers) is set at 0.001:1 to ensure low-temperature printing and limited penetration. In the mass distribution of the polymerizable components, mono-functional monomer should constitute the majority to minimize volume shrinkage. Thus, the mass ratio of the monomer to the total polymerizable components was set at 0.8:1. Furthermore, the appropriate concentrations of the additives STPB and TPP were determined by referring the literature.[2,17] The detailed concentrations of the complete 3D printing formulas are provided in the experiment section.

Next, the optimal formula will be selected based on its excellent photopolymerization capacity (indicated by higher conversion and rate), lower polymerization temperature, and good thermal stability for the actual 3D printing tests. The photopolymerization kinetics of all formulas were tested, and the results are presented in Fig. S3–S6 and Table S3. The experiments were conducted at a relative low light intensity (250 mW cm^{-2}) to ensure that the selected formulas can adapt to the source of actual 3D printing with the same low intensity, as strong light would cause an excessively high temperature. Since cyanines are difficult to dissolve in M2, the relevant tests were discontinued.

$4bZ_3$/Iod as PIS demonstrated outstanding photoinitiation capacity compared to other cyanine/Iod systems (Fig. S3, S4). Its maximum polymerization rates (R_p^{max}) were all above 10 % s^{-1}, and the final conversion (C_f) values of the double bonds all exceeded 70 %. Moreover, when initiating the polymerization of the system containing M1 (Fig. S3a), R_p^{max} reached 39.86 % s^{-1}, indicating that photopolymerization could be nearly accomplished within just a few seconds. The excellent performance of $4bZ_3$/Iod can be explained as follows. Cyanine in the excited state can release energy in three ways, namely, fluorescence (F), vibrational cooling (VC, collision with matrix molecules to release heat) and PET reactions (Scheme 1), a competitive relationship exists among the three.[35,36] $4bZ_3$ exhibited the lowest E_{ox} (0.48 V) and the highest energy of the highest occupied molecular orbital (HOMO) at −2.23 eV among all cyanines (Table S2 and Fig. S7), which is extremely favourable for PET reaction.[2,49,50] Then, $4bZ_3$ possesses a lower free activation enthalpy for PET than other cyanines have, which is also beneficial to PET in ref.[36] In contrast, for the other selected cyanines, a higher free activation enthalpy needs to be overcome.

However, the photoinitiation abilities of these PISs, which are composed of cyanines and OXE-01, are significantly inferior to those of cyanines and Iod. The low E_{red} of OXE-01 (−1.36 V) results in the ΔG_{PET} between cyanines and OXE-01 being positive (Table S2), suggesting that PET has difficulty occurring. Theoretically, for PET between cyanines and oxime esters, additional activation energy derived from heat is



Scheme 1. Simplified process chart of photophysical and photochemical processes occurring within cyanine.

indispensable.[45] Nevertheless, owing to the low light intensity, there is insufficient heat to help them overcome the activation barrier. Furthermore, the lower energy of the lowest unoccupied molecular orbital (LUMO) of Iod (−5.88 eV) also indicates that it has a stronger electron-accepting capacity than OXE-01 (−2.07 eV, Fig. S8). Hence, Iod is a more suitable choice.

Regarding the disparities in polymerization among monomers, specifically the superior polymerization performance of formulas containing M1, their atomic charge distributions were calculated.[41] The results reveal that the double bond of M1 carries a lower negative charge than that of the four monomers (Fig. S9); theoretically, this makes it more prone to react with the *p*-methylphenyl radical generated by Iod, due to the greater difference in charge between them (Fig. S10).

Based on the results of the photopolymerization kinetics, three formulas that concurrently exhibit high C_f and R_p^{max} values (Fig. 2a, b) were selected for the subsequent temperature detection and thermal stability tests. Fig. 2c presents the real-time temperature change during the photopolymerization of the three chosen formulas. The maximum temperature (T_m) reached of the formula containing $5X_4$ was 107.2 °C, far exceeding the T_m of the formula containing $4bZ_3$. The formula containing $4bZ_3$/Iod/M1, which exhibited the best polymerization performance, had the smallest T_m (63.5°C). This is in line with the previous discussion indicating that the more readily the PET takes place, the lower the probability of VC, thereby leading to a reduction of heat released. The thermal stability of these three formulas via DSC (Fig. 2d) were also examined, and the differences among them were insignificant, with all T_i being approximately 100°C. In future experiments, real-time temperature changes during the actual 3D printing process will be monitored, compared them with T_i, and discussed in detail.

The photopolymerization kinetics of the combination of Iod and OXE-01 as co-initiators were also tested, as shown in Fig. S11a. Results demonstrated that the performance of Iod alone was comparable to that when Iod and OXE-01 were used together. Furthermore, the addition of OXE-01 reduced the thermal stability of the formula, with T_i being 95.5 °C (Fig. S11b), suggesting that the use of only Iod as initiator is sufficient.

An analysis of the screening test results clearly revealed that when $4bZ_3$ and Iod are employed as the PIS and M1 is used as the monomer, the 3D printing formula results in the best photopolymerization kinetics performance and a relatively low polymerization temperature. This combination is currently the optimal one. Next, actual 3D printing will be implemented.

3.2. 3D printing tests

The physical structure of SLA 3D printing equipment is displayed in Fig. 3a. As the lifting table descends layer by layer, the objects gradually accumulate and eventually take shape. This is consistent with the principle of traditional SLA 3D printing. Nevertheless, substitution of the

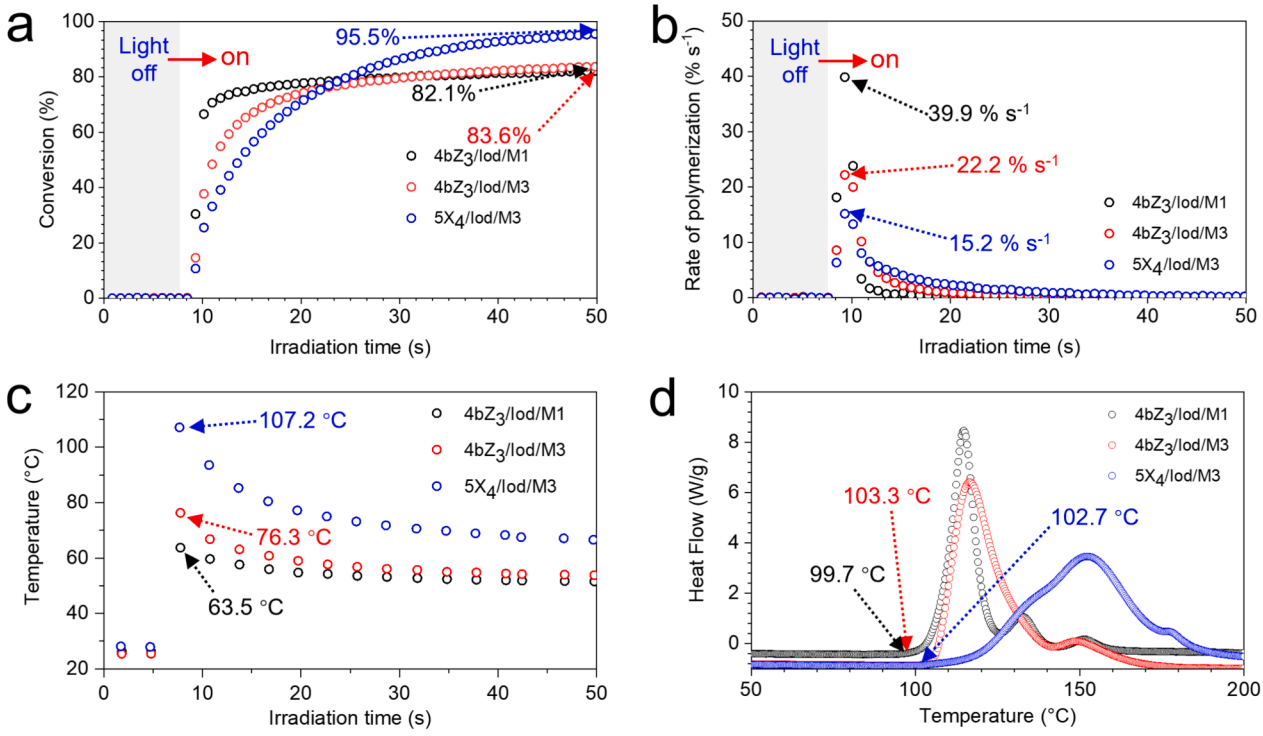


Fig. 2. (a) photopolymerization kinetics and (b) polymerization rates of three chosen 3D printing formulas, which include $4bZ_3$/Iod/M1, $4bZ_3$/Iod/M3, or $5X_4$/Iod/M3 systems, upon irradiation with an 808 nm laser (250 mW cm^{-2}). Curves of (c) polymerization temperature versus irradiation time recorded by a thermal imaging camera; and (d) DSC curves of the three chosen 3D printing formulas, which include $4bZ_3$/Iod/M1, $4bZ_3$/Iod/M3, or $5X_4$/Iod/M3 systems, respectively.

NIR source for the UV source is the main distinction, which represents the innovation of our work. Fig. 3b presents the mechanism of 3D printing with an NIR source. $4bZ_3$ transitions to the excited state upon irradiation with an 808 nm laser and undergoes PET reaction with Iod (also as shown in Eq. (1)). Actually, Eq. (1) is a reversible process. The addition of STPB can, to some extent, promote the forward reaction of Eq. (1), as mentioned earlier. The Iod that subsequently receives electrons ($Iod^{-\bullet}$) undergoes cleavage to produce 4-iodotoluene and *p*-methylphenyl radicals (Eq. (2)), the latter of which are crucial for initiating polymerization.

$$\mathbf{Cyanine}^{*} + \mathbf{Iod} \rightleftarrows \mathbf{Cyanine}^{+\bullet} + \mathbf{Iod}^{-\bullet} \quad (1)$$

$$\mathbf{Iod}^{-\bullet} \rightarrow \mathbf{CH_3Ph}\bullet + \mathbf{CH_3PhI} \quad (2)$$

Eventually, SLA 3D printing using an NIR source was successfully achieved, as presented in Fig. 3c. With a light intensity of 145 mW cm^{-2} and a scanning speed of the laser ranging from 200 to 280 mm s^{-1}, the turnplate, rose, and dogbone shapes for tensile tests were successfully printed, and the printing process can be viewed in supplementary movie 1 and movie 2. In addition, the printed objects replicated the shape of the models (Fig. S12, S13) closely. Therefore, the actual printing experiments demonstrated the feasibility of our strategy.

Next, related chemical and physical properties of the formula used for 3D printing were characterized. Firstly, research on the temperature of the printing process were undertaken; this is because, for NIR photopolymerization, heat is an unavoidable factor, and the photothermal conversion properties of cyanine must be considered. If the temperature is excessively high during the printing process, thermal polymerization will cause serious consequences, such as a reduction in fidelity and even the spread of the polymerization reaction to the entire non-exposed area.

An infrared camera was used for the real-time monitoring of the temperature during the printing process of the dogbone shapes, as shown in supplementary movies 3 and 4. The curves of the temperature changes in the exposed area during the printing of the support and object structure are depicting in Fig. 4a, and b, respectively. About the support structure, there is a periodic temperature change, and each period represents the manufacturing of a single layer. Within the 16 cycles monitored, the T_m was 50.9 °C, which is much lower than the T_i (99.7 °C) detected by DSC. This suggests that the thermal reaction is somewhat difficult to trigger.[2,16] Additionally, the same result was observed through monitoring the printing process of the object structure. While printing 7 layers, the T_m was 76.0 °C. The temperature of the liquid surface in the non-exposed area remained consistently around 36 °C during the processing of the 7-layer object structure (Fig. S21). At the end of the process, the liquid surface had a temperature of 34.7 °C (Fig. 4c), and the object surface reached 41.3 °C (Fig. 4d). In conclusion, considering the heat effect, our printing process is relatively stable, ensuring both printing fidelity and safety.

Fig. 5a shows the steady-state photolysis behaviour of our 3D printing formula when irradiated by an 820 nm LED source. The absorption peak at approximately 800 nm decreases sharply, which is attributed to the cleavage of $[4bZ_3]^{\bullet+}$ produced by the PET between $4bZ_3$ and Iod (Eq (1), resulting in the destruction of the conjugated structure.[36,51] The photolysis rate shown in Fig. S22 further confirms this behaviour, which is embodied in the fact that within just 1 s, the absorption value of the peak decreases by approximately 85 %, and the high photosensitivity is in line with the results of the photopolymerization kinetics. The increase in absorption values within the 400–600 nm range may be attributed to the absorption characteristics of the photolysis products detected by LC-MS (Fig. 5b, Fig. S23–S28). This

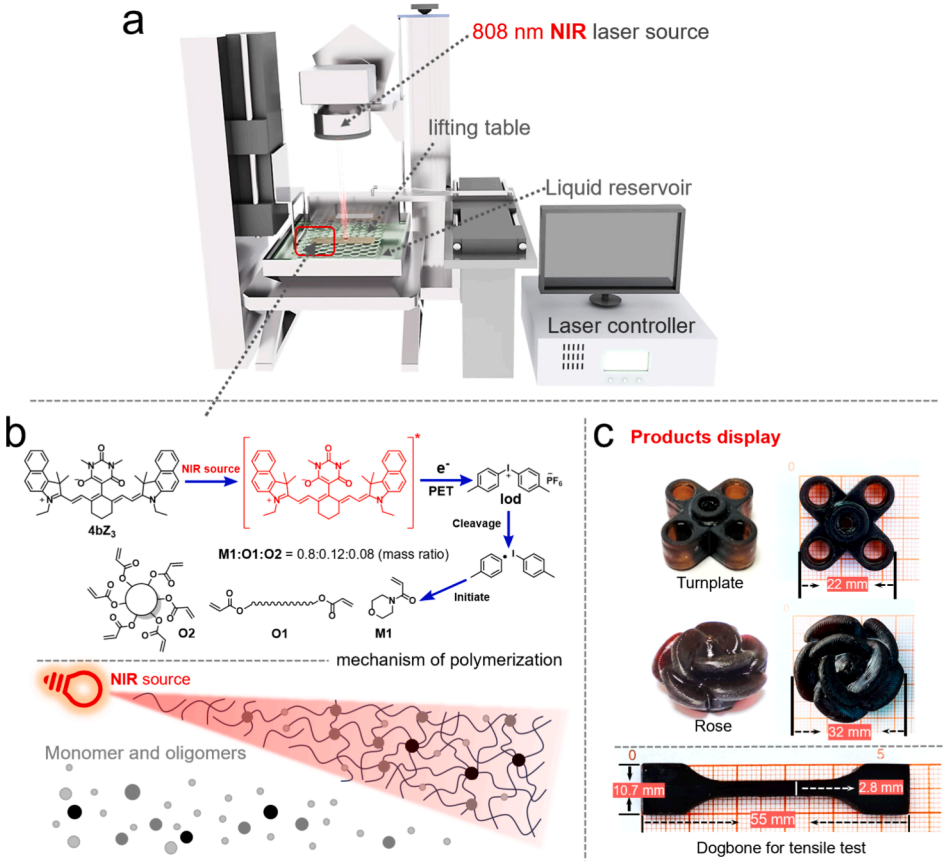


Fig. 3. (a) Physical structure of the SLA 3D printing equipment. (b) Chemical mechanism of SLA 3D printing using 808 nm NIR light as the .
Source. (c) Display of the printed objects, the printing process with a light intensity of 145 mW cm^{-2} and a scanning speed ranging from 200 to 280 mm s^{-1}; formula composed of $4bZ_3$, Iod, STPB, TPP, M1, O1, O2 (0.001:0.02:0.002:0.002:0.8:0.12:0.08, mass ratio)

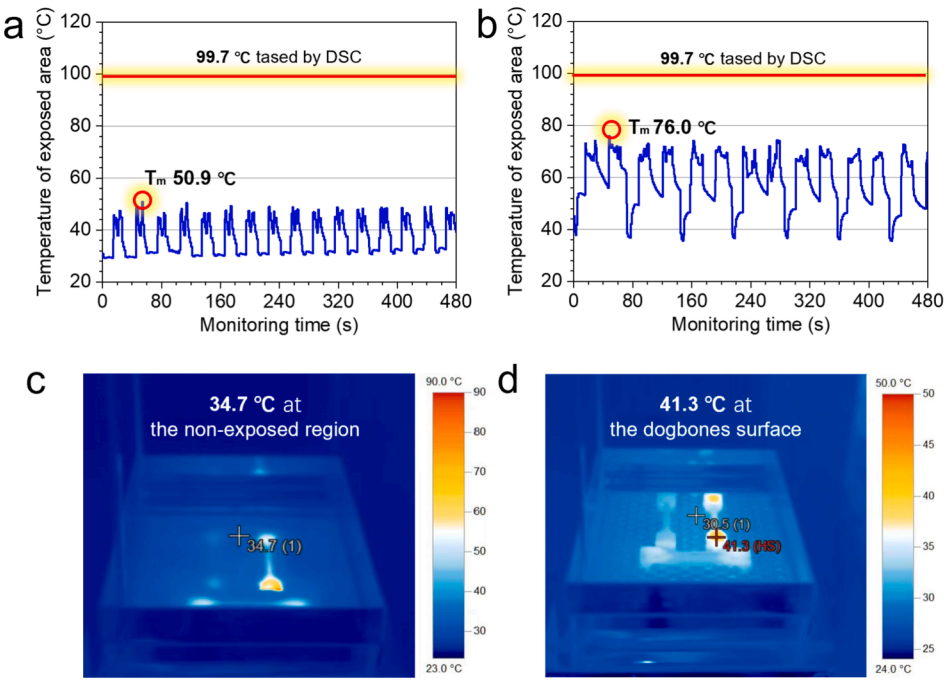


Fig. 4. Curves of the temperature change in the exposed area versus monitoring time during the printing process of the (a) 16-layer support structure and (b) 7-layer object structure of dogbones under a light intensity of 145 mW cm^{-2}. (c) The temperature of the liquid surface (34.7 ℃) in the non-exposed region at the end of the printing process of the dogbones. (d) The temperature of the dogbone surface (41.3 ℃) at the end of the printing process.

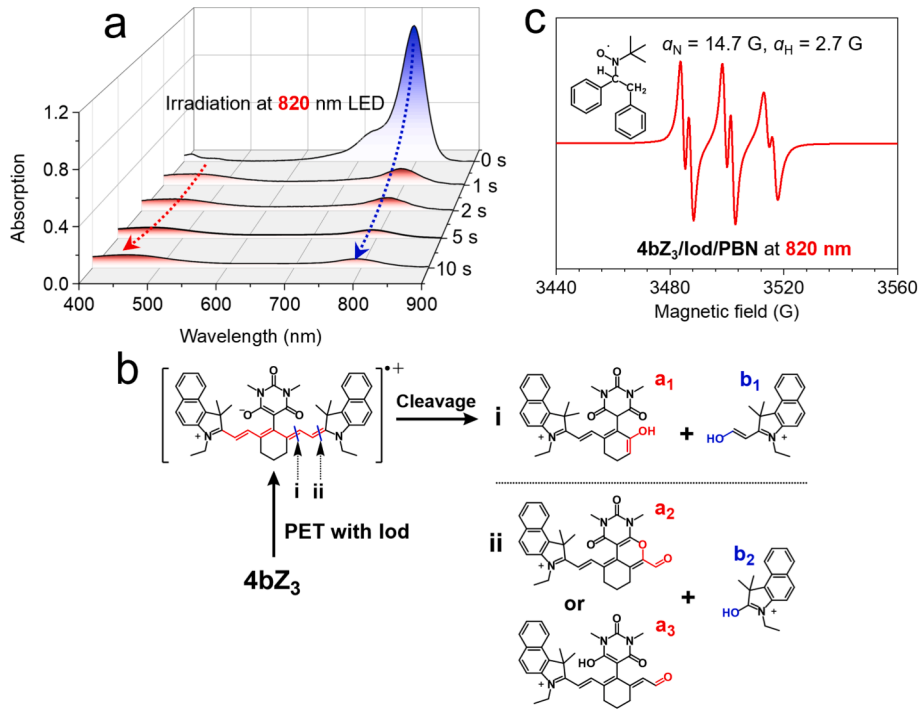


Fig. 5. (a) Steady-state photolysis of the 3D printing formula under irradiation by an 820 nm LED (825 mW cm^{-2}). (b) The cleavage routes of $[4bZ_3]^{\bullet+}$ produced by the PET between $4bZ_3$ and Iod under irradiation at an 820 nm LED (825 mW cm^{-2}, 2 mins), with "i" and "ii"; five photolysis products (a_1, a_2, a_3, b_1, and b_2) were detected via LC-MS in acetonitrile (concentration = 0.5 g L^{-1} for $4bZ_3$, 1 g L^{-1} for Iod). (c) ESR spectra of radicals generated by $4bZ_3$/Iod system under irradiation by an 820 nm LED (825 mW cm^{-2}) trapped by PBN.

explains why the colour changed from green in the original formula to dark brown in the product.

ESR tests showed that the hyperfine coupling constants of the radicals generated by the $4bZ_3$/Iod system in the presence of PBN as a trapping reagent upon exposure to an 820 nm LED were $a_N = 14.7$ G and $a_H = 2.7$ G (Fig. 5c). These values are nearly identical to those of the *p*-methylphenyl radicals generated by Iod under irradiation with a 365 nm LED (a_N = 14.8 G, a_H = 2.7 G, as shown in Fig. S29). Hence, these findings provide evidence that *p*-methylphenyl radicals play a predominant role in initiating the curing of the 3D printing formula.

The mechanical properties of the materials fabricated via 3D printing were investigated by performing tensile tests on the prepared dogbone samples (as shown in Fig. 3c). The three dogbone samples were processed simultaneously and were of the same size (Fig. S30). The test results reveal that these materials can withstand high stresses (Fig. S31), all of which exceed 40 MPa, while the tensile strains are relatively low (approximately 2 %); this suggests that the samples possess high hardness, likely due to the high double bond conversion during polymerization and the use of *hexa*-functional O2, which enhances the degree of crosslinking. Furthermore, the stress – strain curves of the three samples presented nearly identical slopes, demonstrating the uniformity of the 3D printing process.

3.3. Application of SLA 3D printing upon NIR photopolymerization

The products printed using near-UV 3D printing via photopolymerization are predominantly yellow, primarily due to the colour characteristics of the PIS itself.[52,53] Preparing coloured products through near-UV 3D printing is challenging, as coloured fillers usually absorb most of the light, greatly reducing the photoinitiation efficiency of PIs.[2,24,26,54] This is especially the case when black products are manufactured, as black fillers can absorb nearly all the light.[24] Using NIR 3D printing to manufacture coloured products is a promising option, as most coloured fillers have weak absorption in the NIR region.

One issue that arises is the expression of colour. The colour of the printing products mentioned above is dark brown, which can readily obscure the colour of the fillers. To address this problem, lowering the low concentration of $4bZ_3$ is requisite. However, this adjustment introduces a new challenge: NIR light penetrates the system almost unimpeded, which is certain to cause a decline in printing fidelity. A series of corresponding experiments were conducted to illustrate this fact, as depicted in Fig. S32a to e. Formulas with different concentrations of $4bZ_3$ (0.001–0.0002 mass ratio to polymerizable components) and organic dye Rhodamine B (0.005 mass ratio to polymerizable components) were employed to print the letter "N". As the concentration of $4bZ_3$ decreased, the colour of Rhodamine B became more apparent, however, distinct penetration phenomena were observed, leading to the inability to disassemble the objects.

Therefore, the use of a light absorber is inevitable.[55] Its function in the printing formula can be generalized as absorbing excessive light without influencing photopolymerization. Here, cyanine $1aZ_1X_1$, as mentioned above, was utilized as light absorber, attributed to its lower

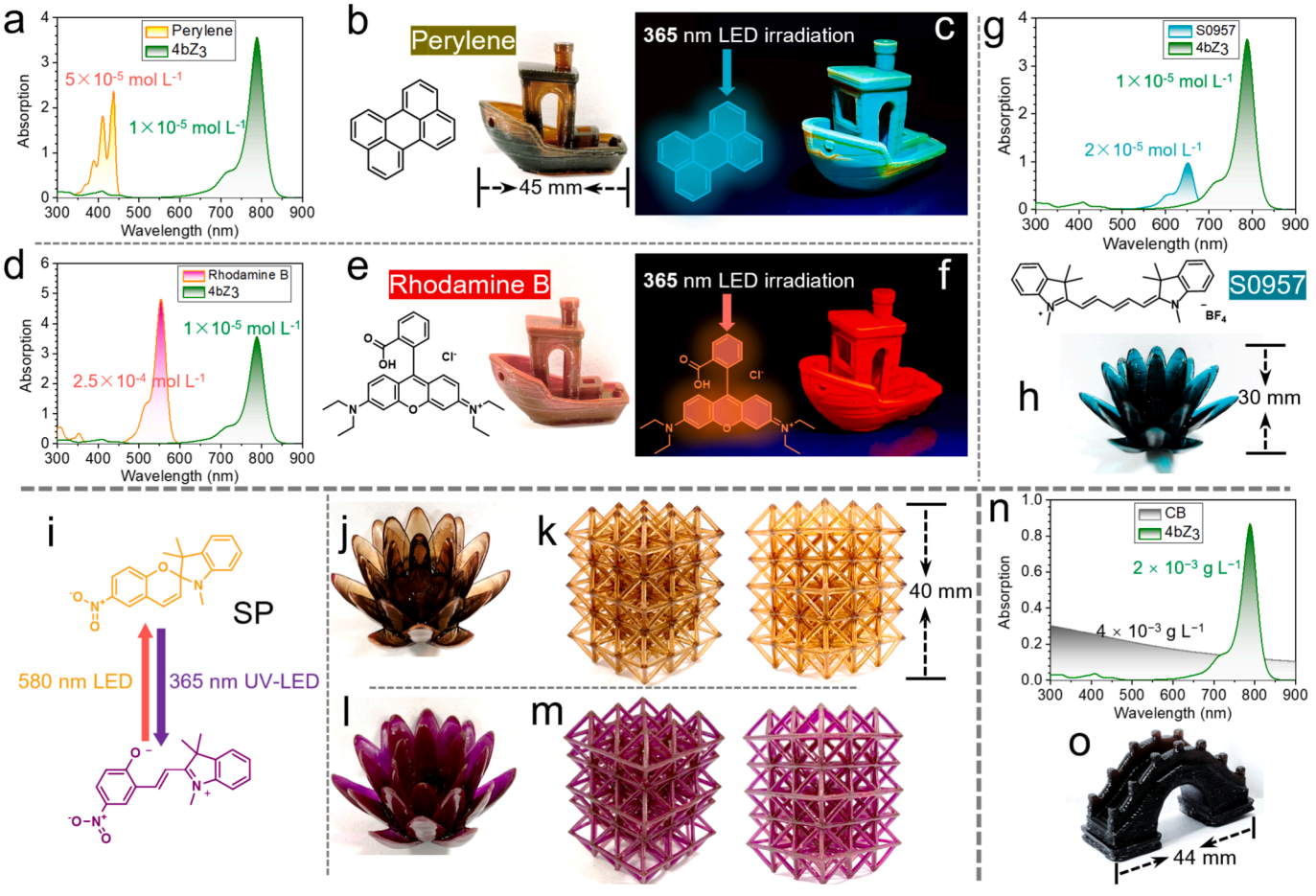


Fig. 6. (a, d) UV–vis-NIR absorption spectra of dichloromethane solutions of perylene (5×10^{-5} mol L^{-1}), Rhodamine B (2.5×10^{-4} mol L^{-1}) and $4bZ_3$ (1×10^{-5} mol L^{-1}). Printed boats containing (b) perylene (5 times the mole of $4bZ_3$) or (e) Rhodamine B (25 times the mole of $4bZ_3$) with a light intensity of 220 mW cm^{-2} and a scanning speed ranging from 100 to 180 mm s^{-1} of laser, and (c, f) fluorescent image under 365 nm LED irradiation. (g) UV–vis-NIR absorption spectra of dichloromethane solutions of S0957 (2×10^{-5} mol L^{-1}) and $4bZ_3$ (1×10^{-5} mol L^{-1}). (h) Printed water lily containing S0957 (2 times the mole of $4bZ_3$) with a light intensity of 220 mW cm^{-2} and a scanning speed ranging from 100 to 180 mm s^{-1} of laser. (i) The photochromic principle of SP under 365 nm and 580 nm LED irradiation. Printed water lily and honeycomb structure containing SP (2 times the mole of $4bZ_3$) under the conditions of (j, k) yellow and (l, m) purple, with a light intensity of 220 mW cm^{-2} and a scanning speed ranging from 100 to 180 mm s^{-1} of laser. (n) UV–vis-NIR absorption spectra of methanol solutions of CB (4×10^{-3} g L^{-1}) and $4bZ_3$ (2×10^{-3} g L^{-1}). (o) Black bridges, printing was achieved with a light intensity of 145 mW cm^{-2} and a scanning speed ranging from 200 to 250 mm s^{-1} of laser, and the mass of CB was 2 times the mass of $4bZ_3$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

light absorption property in the visible light region (without affecting colour expression), outstanding absorption capacity in the NIR region, and relatively weaker photoinitiation ability compared to $4bZ_3$. This ensures that, after absorbing light, it returns to the ground state instead of undergoing a PET reaction and decomposition (Scheme 1).

Fig. S32f to h clearly demonstrates that the addition of $1aZ_1X_1$ effectively hinders the penetration of NIR light while not influencing the colour expression of Rhodamine B. When $1aZ_1X_1$ accounts for 0.0001 of the mass ratios of polymerizable components, no noticeable penetration phenomena are observed during printing. Consequently, the mass ratio of $4bZ_3$ was reduced to 0.0002, and an additional 0.0001 mass ratio of $1aZ_1X_1$ was added to obtain the new formula for the subsequent 3D printing with coloured fillers, while the mass ratio of the other components remains unchanged. The organic dyes perylene, Rhodamine B, and pentamethine cyanine S0957 were added to the printing formula for the manufacture of polychromatic products. Their absorption peaks are located within the ranges of 400–500 nm (Fig. 6a), 500–600 nm (Fig. 6d), and 600–700 nm (Fig. 6g) respectively, nearly covering the entire visible light region. Moreover, their absorption are precisely staggered with respect to $4bZ_3$. That is, the organic dyes are responsible for expressing the colour of the products, while $4bZ_3$ is utilized to initiate polymerization for printing.

The results of photopolymerization kinetics suggest that compared to the original formula shown in Fig. 3, the new formulas with a low content of $4bZ_3$ and containing organic dyes exhibit lower C_f and polymerization rates, but they can still polymerize effectively (Fig. S34). By raising the light intensity to 220 mW cm^{-2} and lowering the laser scanning speed to 100–180 mm s^{-1} during the printing process, the reduction in photopolymerization capability can be effectively compensated. Finally, the boats containing perylene (Fig. 6b) and Rhodamine B (Fig. 6e), as well as the water lily containing S0957 (Fig. 6h), were successfully manufactured. The organic dyes endowed them with new colours, replacing the brown of the original formula shown in Fig. 3. Furthermore, due to the fluorescence properties of perylene and Rhodamine B, the boats presented a luminous effect under the irradiation of a near-UV source (Fig. 6c, f), resembling the image of sailing on the sea at night.

Meanwhile, a temperature monitoring experiment of the exposure area during the printing process of the boat containing Rhodamine B was conducted. The T_i of the formula tested through DSC was 98.3 ℃ (Fig. S35b), which is higher than the T_m (69.7 ℃) observed during the printing process (Fig. S35a), signifying that the printing process was thermally stable.

Photochromic products were manufactured by adding SP to the new formula.[56] The photochromic mechanism is depicted in Fig. 6i. The purple and yellow states can be reversibly switched by light irradiation, and the corresponding changes of absorption curves are shown in Fig. S36. Photochromic water lily (Fig. 6j, l) and honeycomb structures (Fig. 6k, m) were successfully printed, and the manufacture of the honeycomb indicates that the strategy of NIR 3D printing for preparing hollowed-out structures is feasible.

The fabrication of black products was carried out by adding carbon black (CB) to the formula. Fabricating these products is challenging through traditional UV photopolymerization as black fillers absorb nearly all the light, thereby hindering the polymerization. Fig. 6n presents a comparison of the absorption capabilities of CB and $4bZ_3$. Although CB has a considerable absorption value in the NIR region, it is still unable to compete with $4bZ_3$. Since the colour expression is not a concern in this case, it is feasible to use the original formula with a higher content of $4bZ_3$ (mass ratio of 0.001 to polymerizable components) without addition of the $1aZ_1X_1$. Meanwhile, the results of photopolymerization kinetics also suggest that the addition of carbon black has a very minor impact on polymerization (Fig. S34). Based on this, a black bridge was printed (Fig. 6o). The ability to fabricate black products is of great significance for 3D printing via photopolymerization. The SLA 3D printing approach via NIR photopolymerization is a novel method for processing products with colour, including stimulus-responsive materials capable of photochromism.

3.4. Evaluation of printing fidelity

Errors are unavoidable between the 3D printing model and the actual object due to factors such as the light penetration and scattering. Nevertheless, the printing fidelity can be enhanced through various approaches, such as adding light absorbers to the formula to mitigate these effect, which has been mentioned in the previous section.[55] Indeed, $1aZ_1X_1$ can reduce unnecessary light penetration. Meanwhile, the CB is another viable option and has been reported to be effective.[32].

To evaluate the error range of our NIR SLA printing method and the feasibility of enhancing the precision of $1aZ_1X_1$ and CB, the following series of experiments were designed. The fine 3D model depicted in Fig. S37a was designed, featuring 8 cuboids with widths of 1 mm arranged in parallel, and with distances between them ranging from 0.5 mm to 3.5 mm. The printed objects without (formula same as Fig. 3), and with $1aZ_1X_1$ or CB are presented in Fig. S37b. Due to the scattering effect and diffusion of polymerization, cuboids #1 and #2 adhered. Optical microscopy revealed that the width of cuboid #8 without $1aZ_1X_1$ or CB is 1178 μm (Fig. S38a, theoretically 1000 μm); thus, the error is approximately 0.2 mm. However, the width of cuboid #8 with $1aZ_1X_1$ added is 1118 μm (Fig. S38d), and with CB is 1045 μm (Fig. S38g), indicating that the increase in precision caused by the addition of $1aZ_1X_1$ or CB is remarkable, especially for the latter. The same conclusion can be drawn by comparing the widths of cuboids #7, and the spacings between cuboids #4 and #5 (Fig. S38b, c, e, f, h, i).

Fig. S19, 7a, d, and g present the 3D model and the actual printed products of the bracelet with hollowed-out structures. The success of our NIR 3D printing strategy in printing more challenging hollowed-out structures is readily observable. Nevertheless, in the absence of $1aZ_1X_1$ or CB, the fidelity of the printed product is poorer than that of the model when $1aZ_1X_1$ or CB is added, as reflected in the slope structures shown in Fig. 7b, e, and h, as well as the vertical structures in Fig. 7c, f and i. Owing to the rapid photolysis of $4bZ_3$, light can effortlessly penetrate the interior of the system, resulting in the thickening of the slope structure (theoretically 1000 μm, actually 2067 μm without $1aZ_1X_1$ or CB). However, the addition of $1aZ_1X_1$ decreases the thickness to 1478 μm, CB decreases the thickness to 1229 μm. With respect to the vertical structures (theoretically 1000 μm), the thickness after the addition of $1aZ_1X_1$ is 1207 μm, CB is only 1107 μm, particularly after the addition of CB, it is far superior to the 1561 μm of the original formula.

Subsequently, we attempted to print a more complex object featuring hollowed-out structures, as depicted in Fig. 7j, k, l, and m. Within a cube with a three-dimensional size of merely 20 mm, regardless of whether $1aZ_1X_1$ or CB was added, the hollowed-out structure maintained excellent fidelity compared to the 3D model (Fig. S20). These results represent an improvement over existing NIR 3D printing methods. After all, based on current reports, manufacturing similar products presents certain difficulties.

Above outcomes indicate that the effect of $1aZ_1X_1$ and CB as light absorbers in enhancing printing accuracy is evident. Additionally, the utilization of CB achieves two goals simultaneously. On the one hand, it successfully manufactures black products, which is difficult to accomplish through traditional near-UV 3D printing. On the other hand, CB can reduce the printing error within a small and acceptable range; this represents a groundbreaking advancement in processing complex products featuring hollow structures via NIR 3D printing.

4. Conclusions

The SLA 3D printing approach based on NIR photopolymerization was successfully implemented. Among all the designed 3D printing formulas, the formula using $4bZ_3$/Iod as the PIS and M1 as the monomer

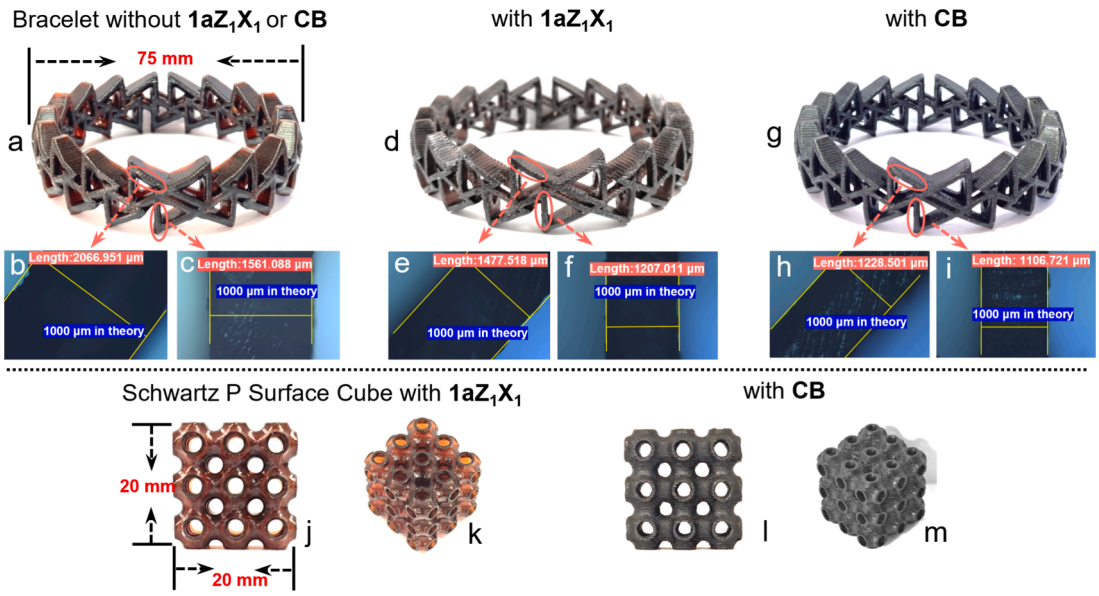


Fig. 7. The bracelets printed by three formulas, (a) without, (d) with $1aZ_1X_1$ (0.2 times the mass of $4bZ_3$), and (g) with CB (2 times the mass of $4bZ_3$); The comparisons of the widths at the same positions by optical microscopy at a magnification of 5×, including the (b, e, h) slope and the (c, f, i) vertical structures, are shown; Printed Schwartz P Surface Cube by formulas (j, k) with $1aZ_1X_1$ (0.2 times the mass of $4bZ_3$), and (l, m) with CB (2 times the mass of $4bZ_3$).

proved to be the optimal choice, due to its high double-bond C_f, high polymerization rate, and relatively low polymerization reaction temperature. This is attributed to the fact that the PET reaction between $4bZ_3$ and Iod occurs more readily than it does in other systems. Consequently, actual 3D printing tests were carried out, and various products were manufactured. The heat effect of NIR 3D printing was also investigated in detail, and the printing process was found to be thermally stable. Furthermore, due to the excellent NIR absorption characteristics of $4bZ_3$, polychromatic products—especially black products—were successfully produced by adding coloured fillers to the 3D printing formula. This is because coloured fillers have difficulty competing with $4bZ_3$ for the absorption of NIR light. Achieving this, however, is difficult through traditional near-UV 3D printing. Meanwhile, stimulus-responsive materials capable of photochromism were also fabricated by adding spiropyran derivatives to the formula. Finally, by adding CB to the 3D printing formula, both the production of black products and the enhancement of printing precision were achieved simultaneously. The printing error was even reduced to approximately 0.1 mm, effectively "killing two birds with one stone". This study expands the range of 3D printing methods based on photopolymerization and achieves significant advancements in the fabrication of delicate structures via NIR 3D printing. Moreover, it broadens the application scope of NIR photopolymerization technology, particularly that based on cyanine.

CRediT authorship contribution statement

Xianglong He: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Yayu Shao:** Software, Methodology. **Yangyang Xin:** Resources, Methodology, Investigation. **Yulian Pang:** Resources, Methodology, Investigation. **Zehao Jin:** Methodology. **Dongyang Guo:** Methodology. **Yingquan Zou:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.160857.

Data availability

Data will be made available on request.

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